

Intro to Machine Learning: Linear Regression

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Content

- What is Machine Learning?
- Supervised Learning
- Unsupervised Learning
- Linear Regression with One Variable
- Linear Regression with Multiple Variables

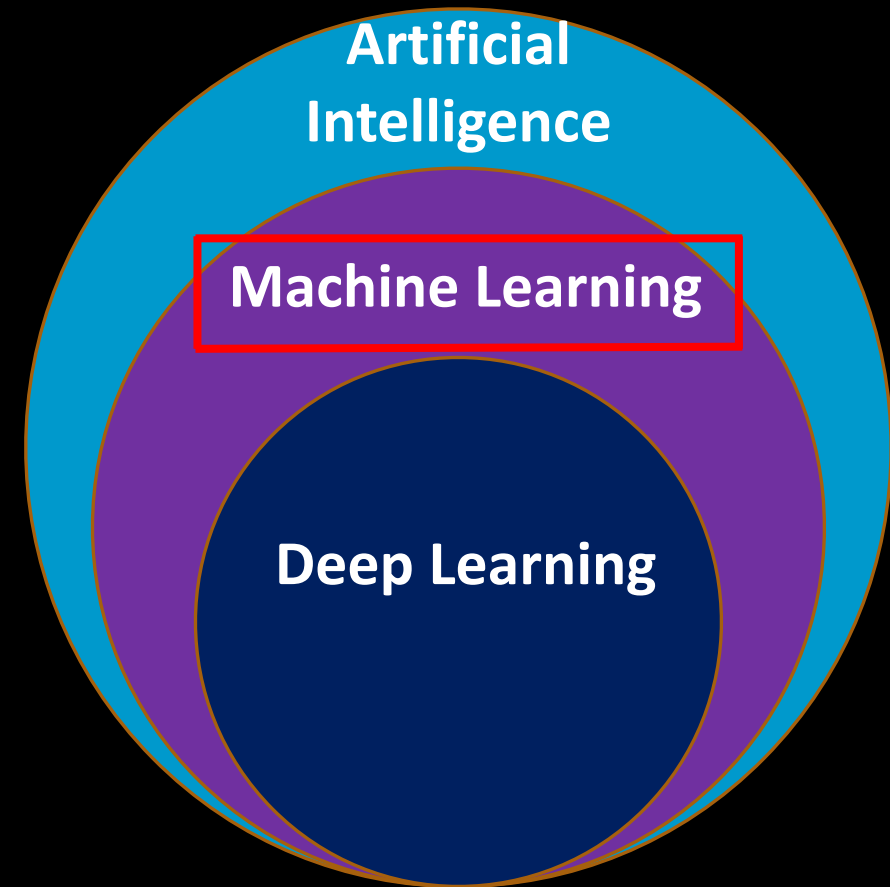
What is Machine Learning?

Machine Learning (ML) – the use and development of computer systems that are able to learn and adapt without following explicit instructions, by using algorithms and statistical models to analyse and draw inferences from patterns in data.

Machine Learning (ML) – Field of study that gives computers the ability to learn without being explicitly programmed. Arthur Samuel (1959).

What is Machine Learning?

Machine Learning is a part of Artificial Intelligence.



What is Machine Learning?

Machine learning

Supervised learning

Unsupervised learning

Reinforcement learning

Recommender system

Content

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Supervised Learning

Supervised learning



A. Classification problem

- Logistic Regression
- Decision Tree
- Naive Bayes
- K – Nearest Neighbor
- Support Vector Machine

Example:

- Email spam detection
- Speech Recognition

B. Regression problem

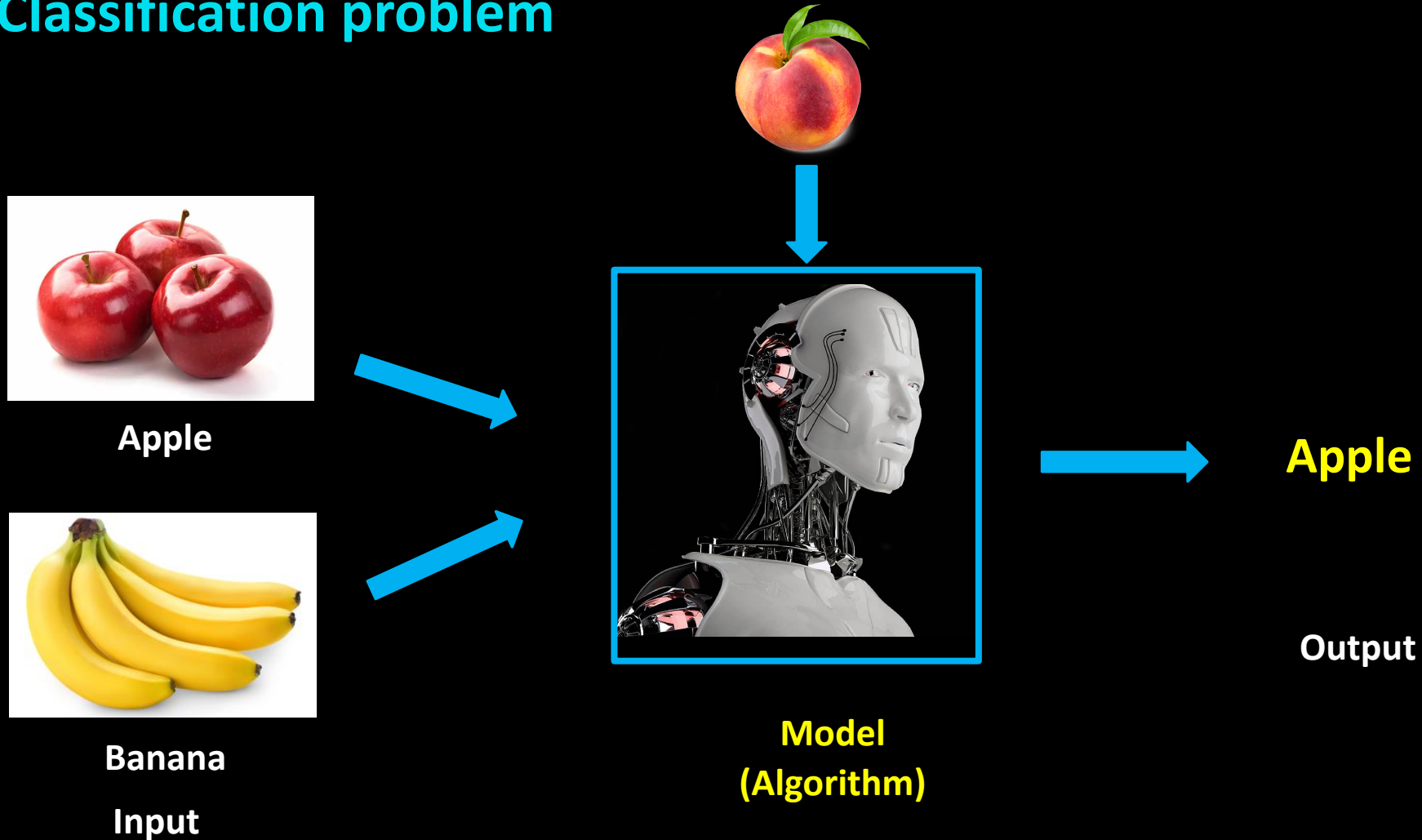
- Linear Regression
- Ridge Regression
- Stepwise Regression

Example:

- Stock Market Prediction
- Rainfall prediction

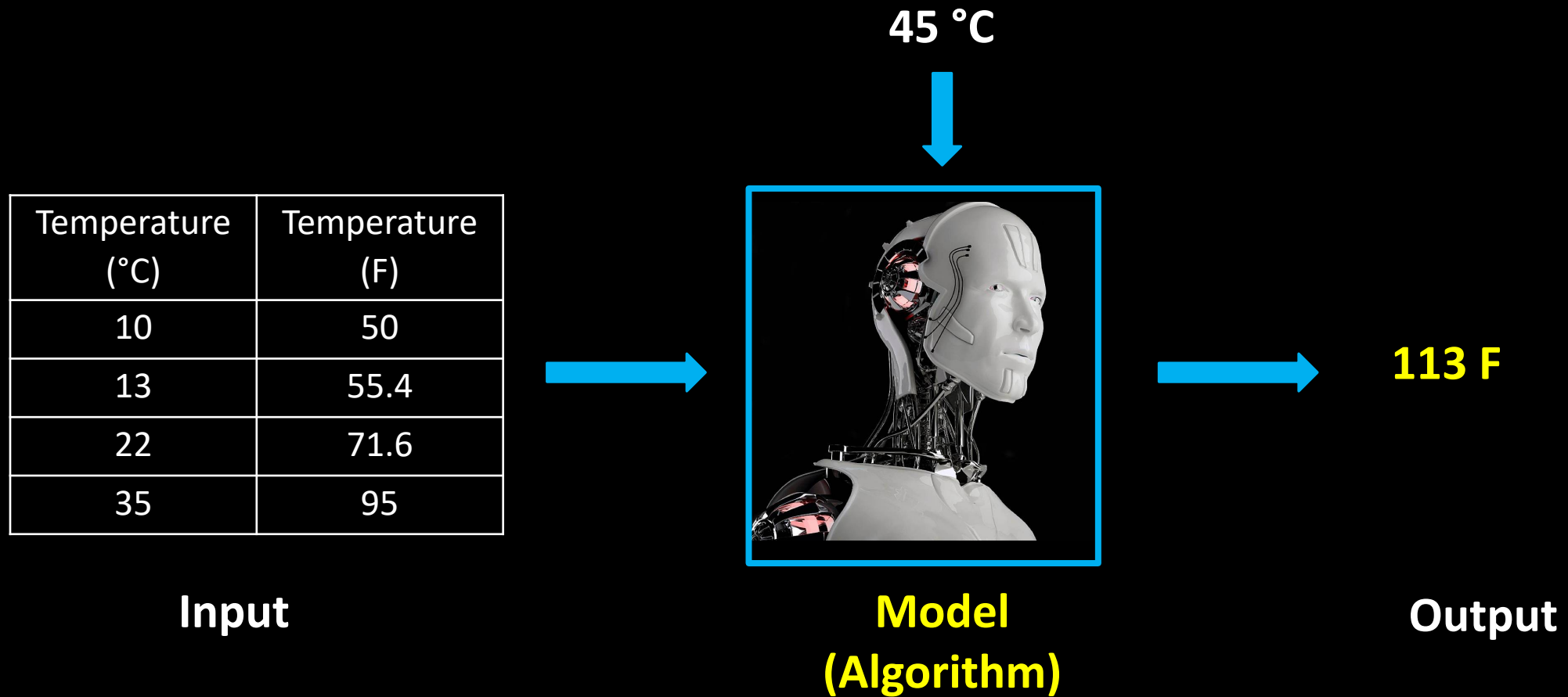
Supervised Learning

Classification problem



Supervised Learning

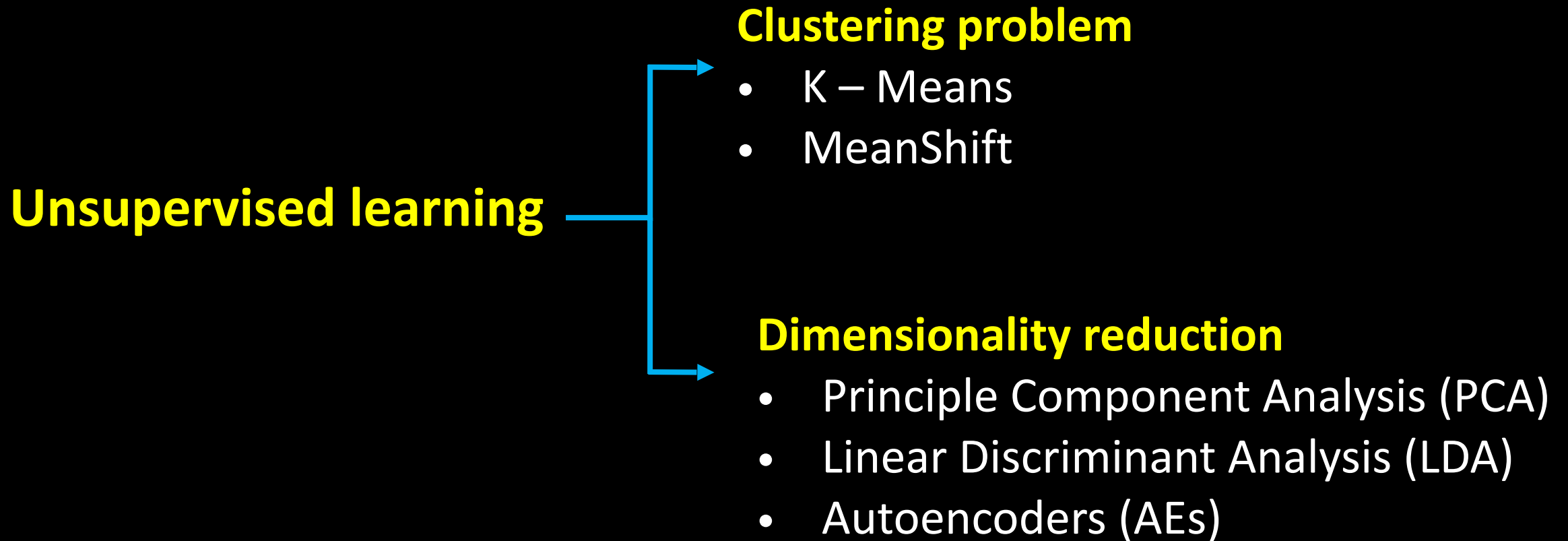
Regression problem



Content

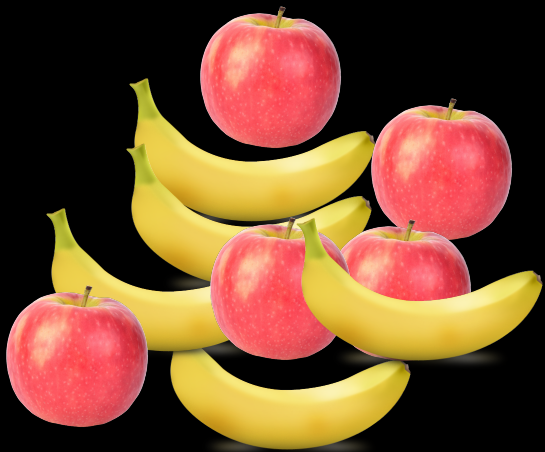
- ~~What is Machine Learning?~~
- ~~Supervised Learning~~
- ~~Unsupervised Learning~~
- ~~Linear Regression with One Variable~~
- ~~Linear Regression with Multiple Variables~~

Unsupervised Learning

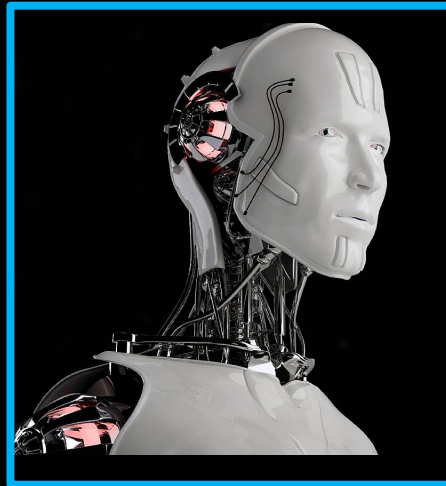


Unsupervised Learning

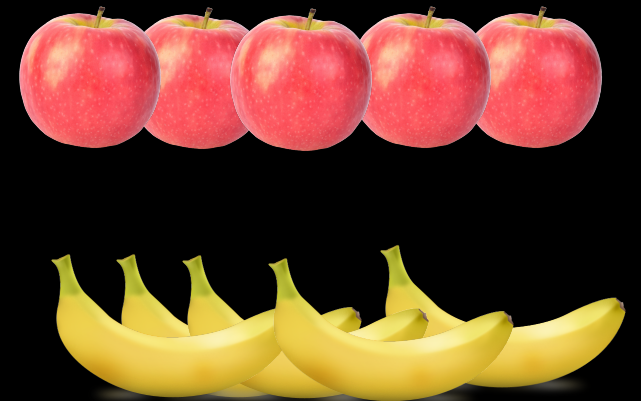
Clustering problem



Input



Model
(Algorithm)



Output

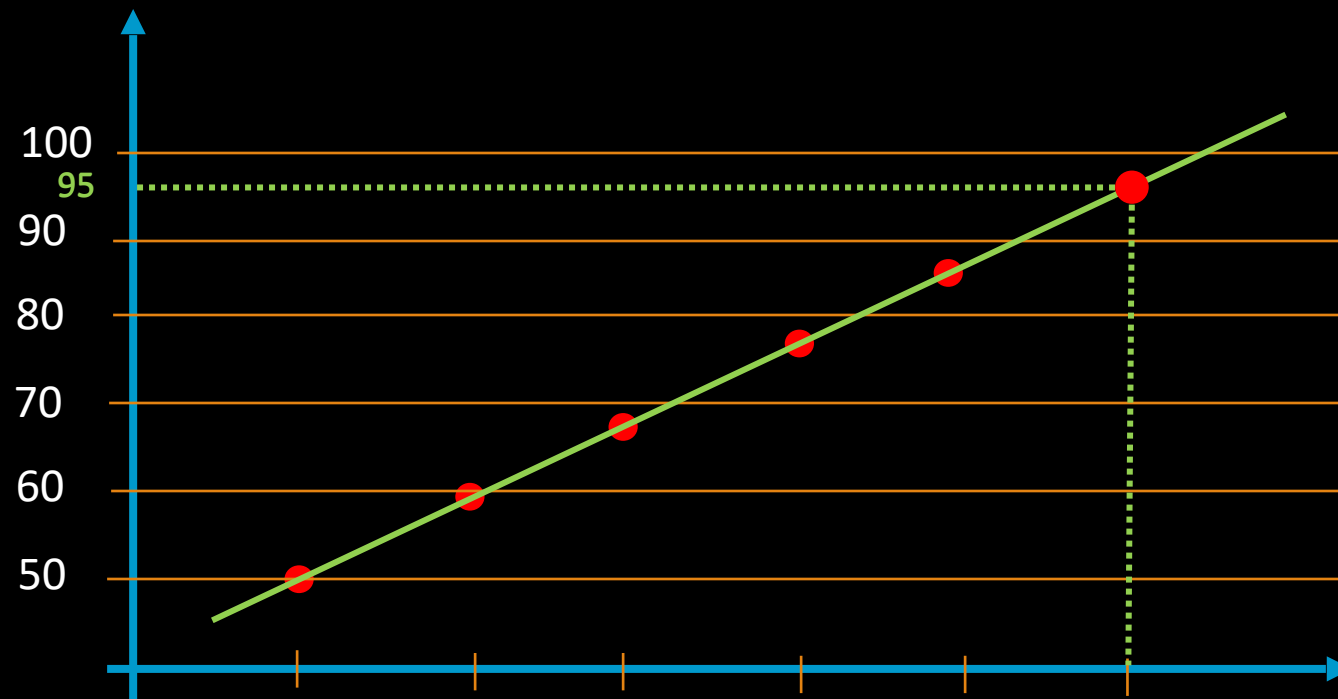
Content

- ~~What is Machine Learning?~~
- ~~Supervised Learning~~
- ~~Unsupervised Learning~~
- Linear Regression with One Variable
- Linear Regression with Multiple Variables

Dataset

Temp (°C) (x)	Temp (F) (y)
10	50
15	59
20	68
25	77
30	86
35	

$$F = 32 + 1.8 \cdot t$$



Houses prices (Dataset)

Size (feet ²), (x)	Prices (in 1000 of \$) (y)
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2104	460
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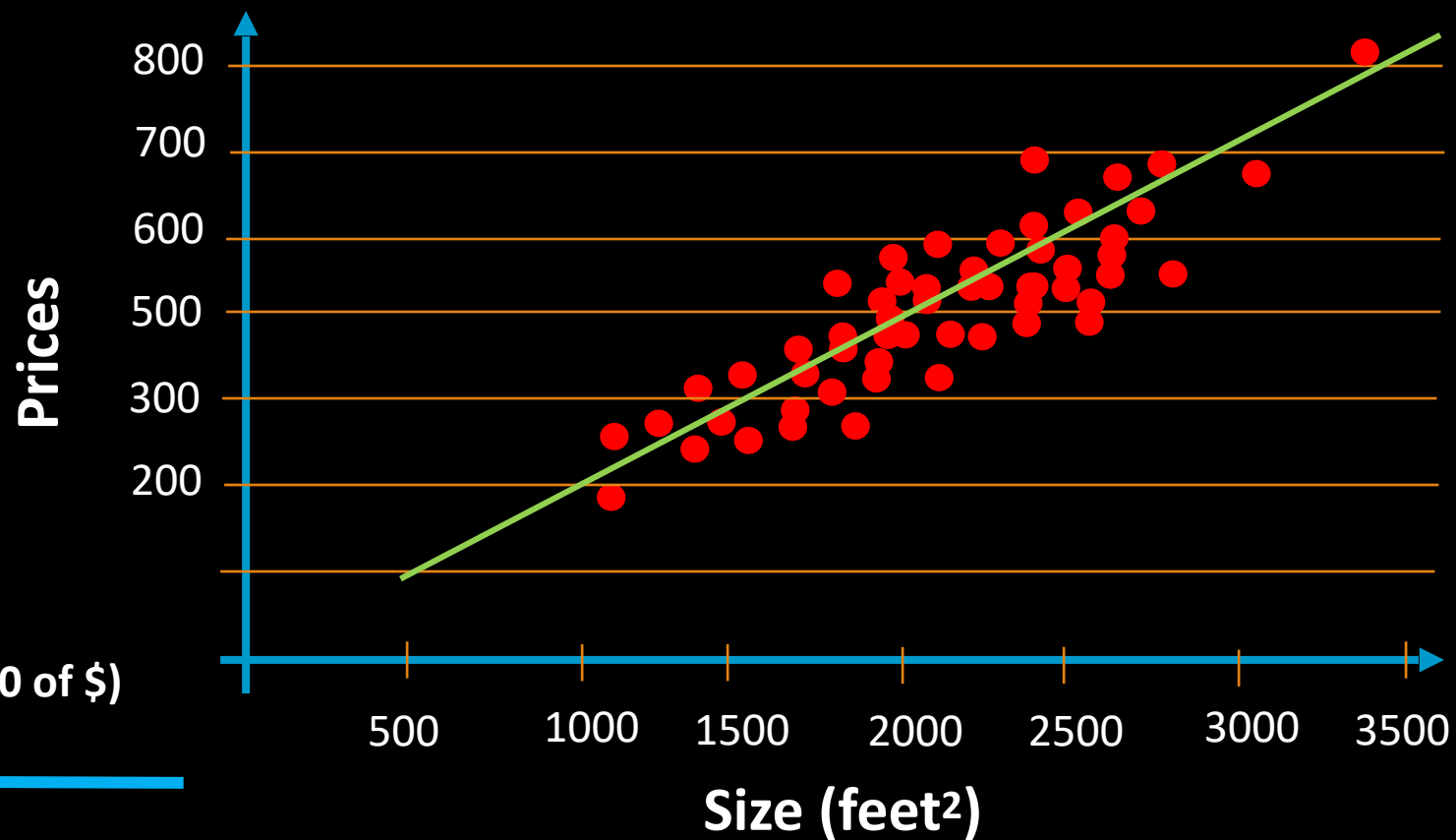
1416	232
------	-----

1534	315
------	-----

852	178
-----	-----

...

...



Training set of Temp. Dataset

m { 1
2
3
4
...

x { 10
15
20
25
...

y { 50
59
68
77
...

Temp (C°), (x)

Temp F (y)

x = “input” variable/feature

y = “output” variable/target

m = Number of training examples

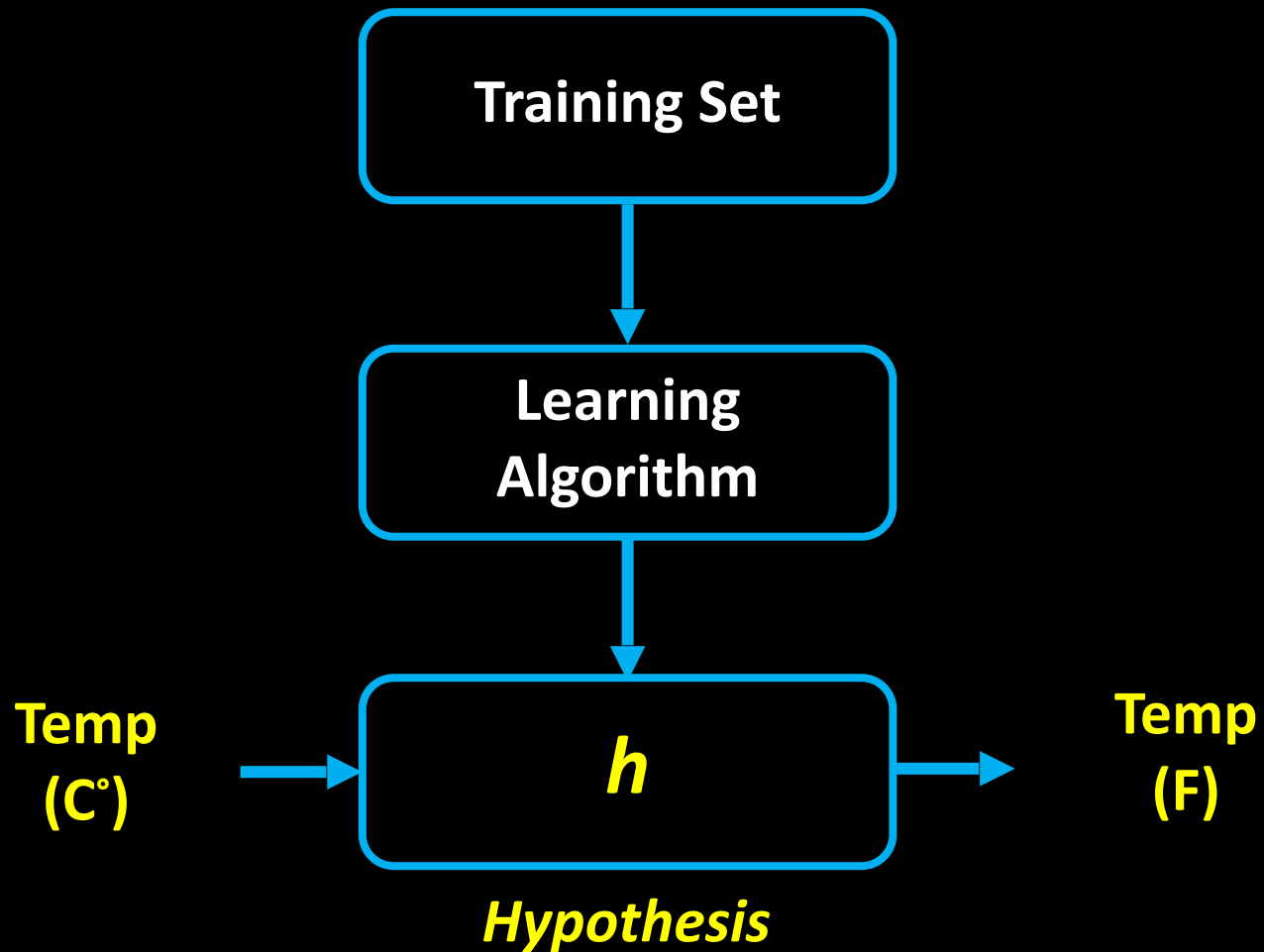
(x, y) – one training example

$(x^{(i)}, y^{(i)})$ – i th training example

$$(x^{(1)}, y^{(1)}) = (\quad , \quad)$$

$$(x^{(2)}, y^{(2)}) = (\quad , \quad)$$

$$(x^{(3)}, y^{(3)}) = (\quad , \quad)$$



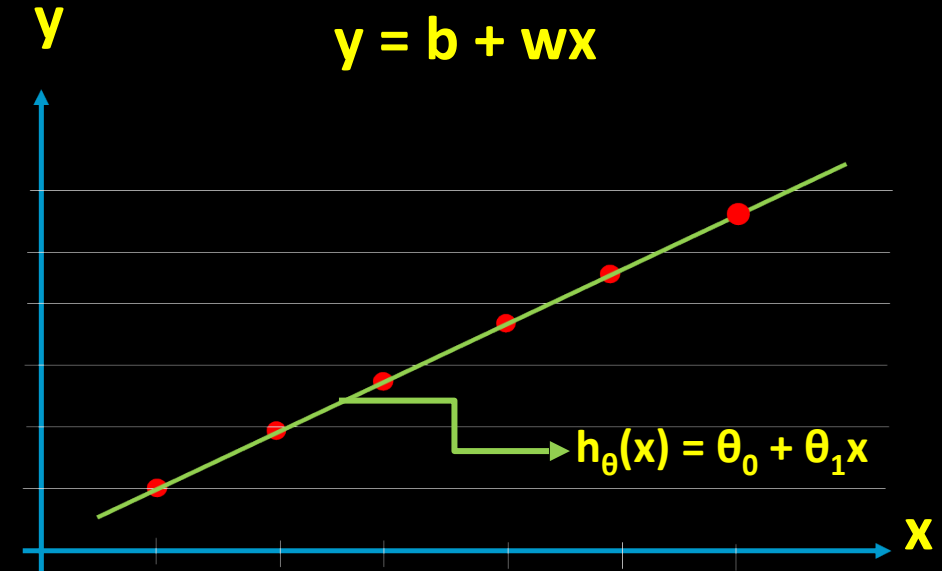
$$y = h_{\theta}(x) =$$
$$\theta_0 = b =$$

$$\theta_1 = w =$$
$$x =$$

How do we represent h ?

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

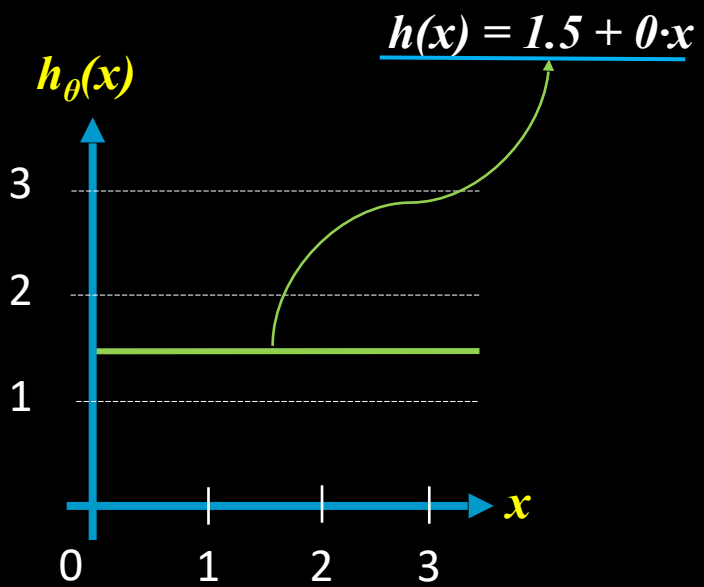
$$y = b + wx$$



$$F = 32 + 1.8 \cdot t$$

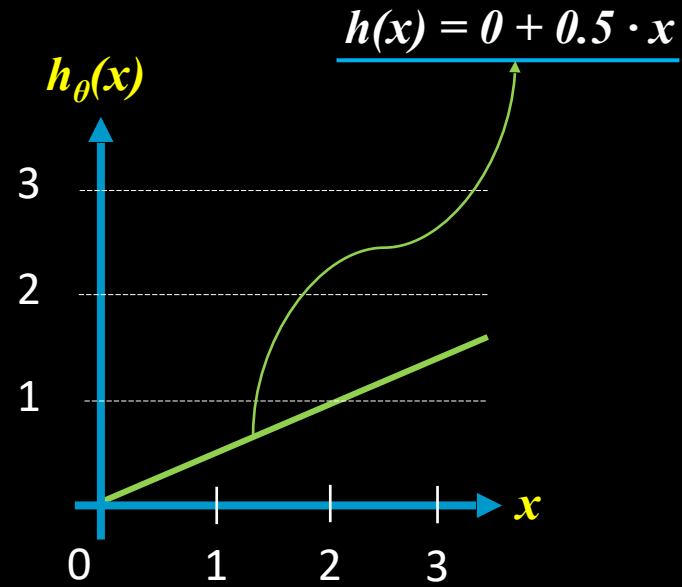
Linear Regression with one variable

Hypothesis: $h_{\theta}(x) = \theta_0 + \theta_1 x$



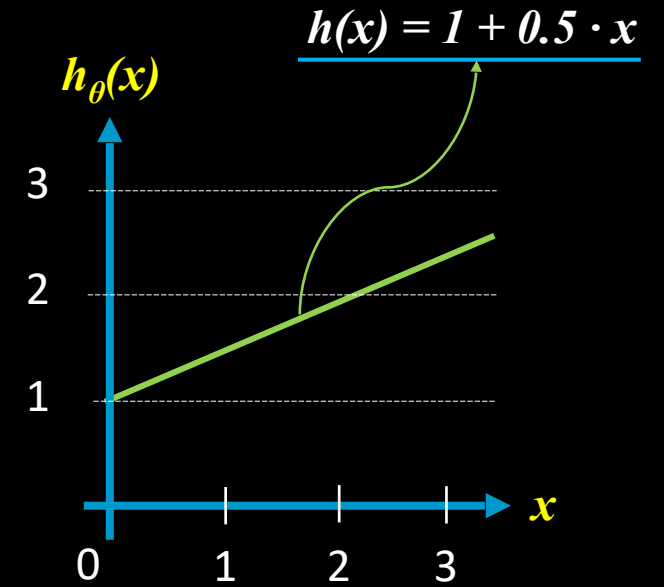
$$\theta_0 = 1.5$$

$$\theta_1 = 0$$



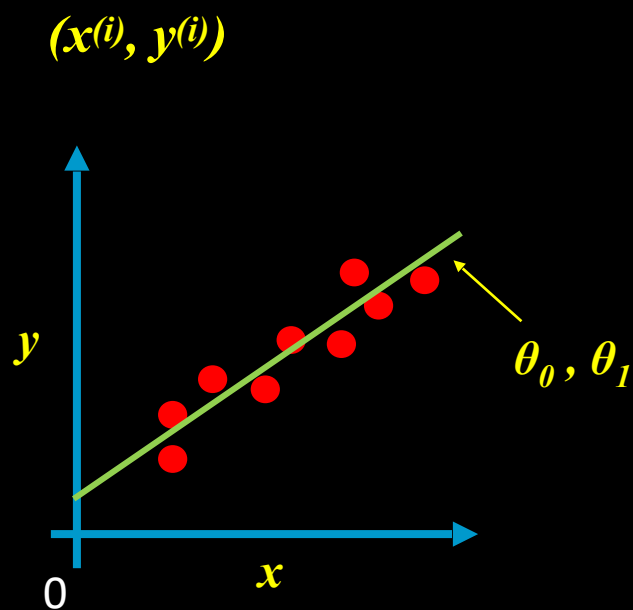
$$\theta_0 = 0$$

$$\theta_1 = 0.5$$



$$\theta_0 = 1$$

$$\theta_1 = 0.5$$



Idea: Choose θ_0, θ_1 so that $h_{\theta}(x)$ is close to y for our training examples (x, y) .

$$\underset{\theta_0, \theta_1}{\text{minimize}} \quad \frac{1}{2m} \sum_{i=1}^m \left(\underbrace{h_{\theta}(x^{(i)})}_{\substack{\downarrow \\ h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}}} - y^{(i)} \right)^2$$

$$h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2 \rightarrow$$

**Cost
function**

$$\underset{\theta_0, \theta_1}{\text{minimize}} \quad J(\theta_0, \theta_1)$$

Squared error function

Hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Parameters:

$$\theta_0, \theta_1$$

Cost function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

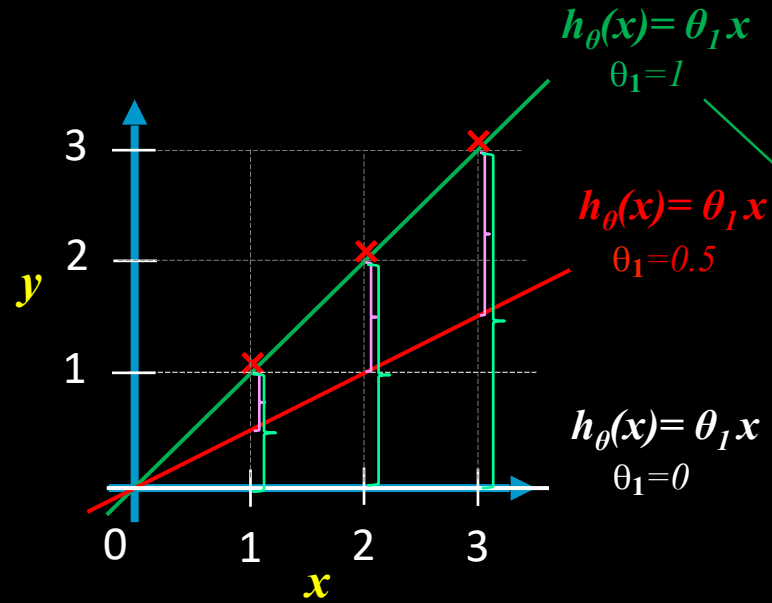
Goal:

$$\underset{\theta_0, \theta_1}{\text{minimize}} \quad J(\theta_0, \theta_1)$$

Simplified:

$$= 0$$

$$J(\theta_1) =$$

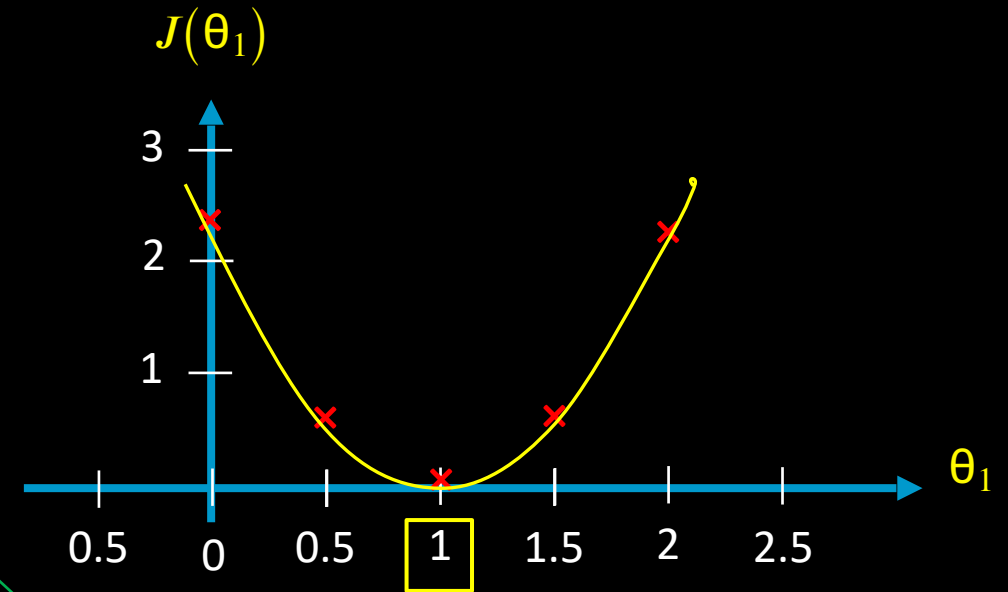
$h_{\theta}(x)$ 

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$J(1) = \frac{1}{2m} ((1 - 1)^2 + (2 - 2)^2 + (3 - 3)^2) = \frac{1}{2 \cdot 3} \cdot 0 = 0$$

$$J(0.5) = \frac{1}{2m} ((0.5 - 1)^2 + (1 - 2)^2 + (1.5 - 3)^2) = \frac{1}{2 \cdot 3} \cdot 3.5 = 0.58$$

$$J(0) = \frac{1}{2m} ((0 - 1)^2 + (0 - 2)^2 + (0 - 3)^2) = \frac{1}{2 \cdot 3} \cdot 14 = 2.33$$

 $J(\theta_1)$ 

θ_1	1	0.5	0	1.5	2
$J(\theta_1)$	0	0.58	2.33	0.58	2.33

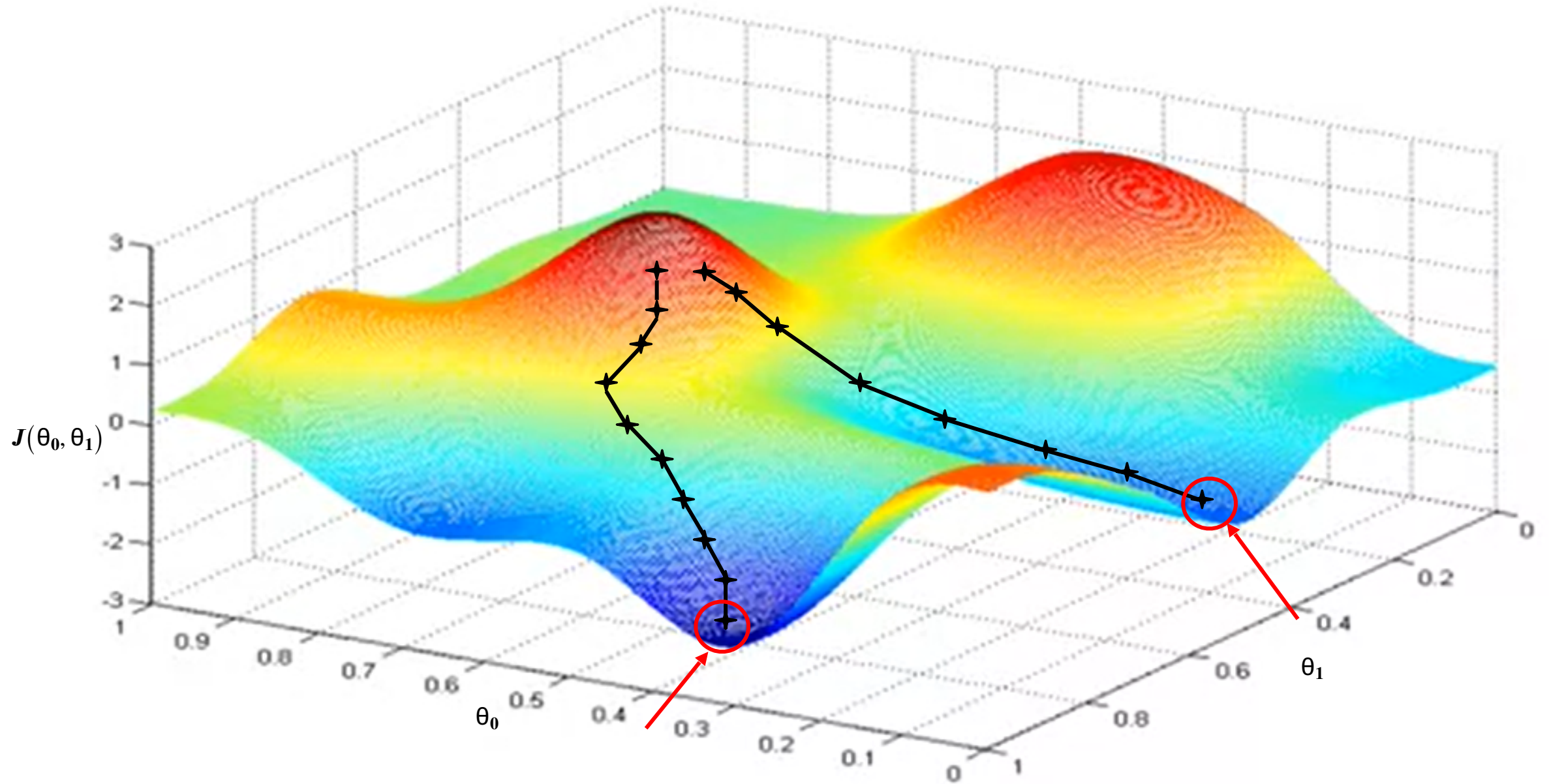
minimize $J(\theta_1)$
 θ_1

Cost function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\Theta}(x^{(i)}) - y^{(i)})^2$

Goal: minimize $J(\theta_0, \theta_1)$
 θ_0, θ_1

Outline:

- Start with some (θ_0, θ_1) $\theta_0 = 0, \theta_1 = 0$
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we hopefully end up at a minimum.



Gradient descent algorithm

Repeat until convergence {

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

(for $j = 0$ and $j = 1$)

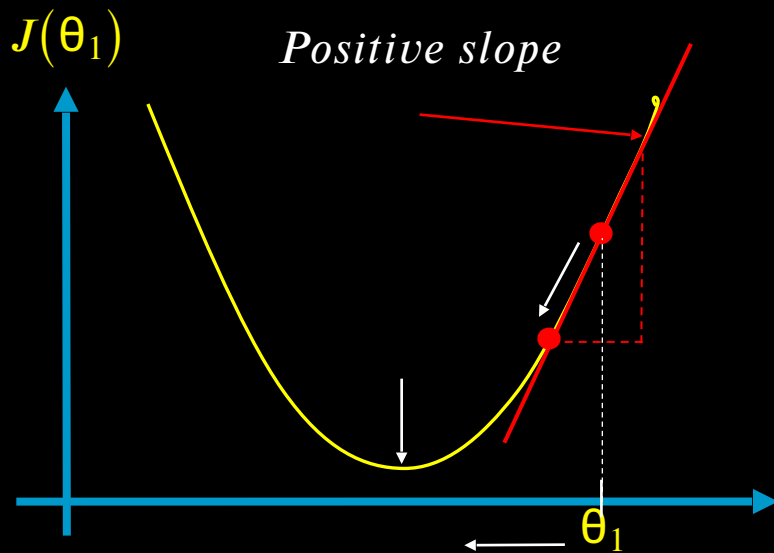
}

Learning rate

Derivative part

minimize $J(\theta_1)$
 θ_1

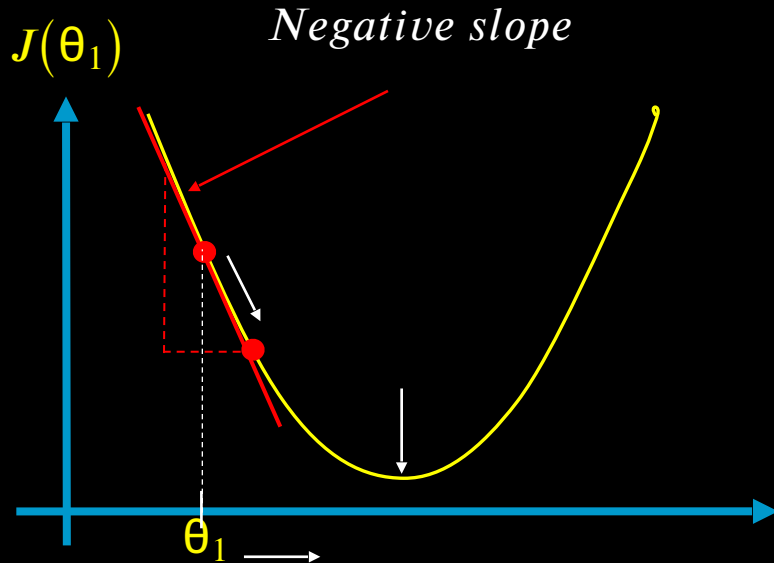
$\theta_1 \in \mathbb{R}$



$$\theta_1 = \theta_1 - \alpha \cdot \boxed{\frac{\partial}{\partial \theta_1} J(\theta_1)}$$

$$\geq 0$$

$$\theta_1 = \theta_1 - \alpha \cdot (\text{Positive value})$$



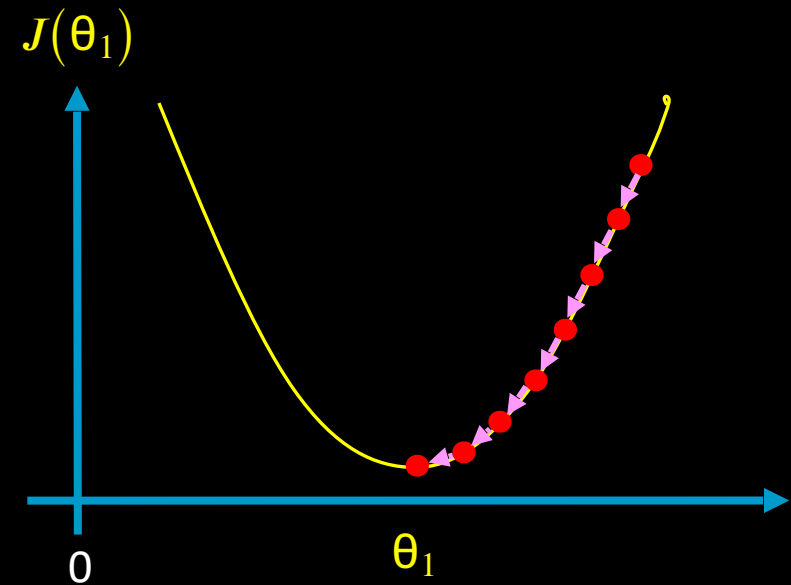
$$\theta_1 = \theta_1 - \alpha \cdot \boxed{\frac{\partial}{\partial \theta_1} J(\theta_1)}$$

$$\leq 0$$

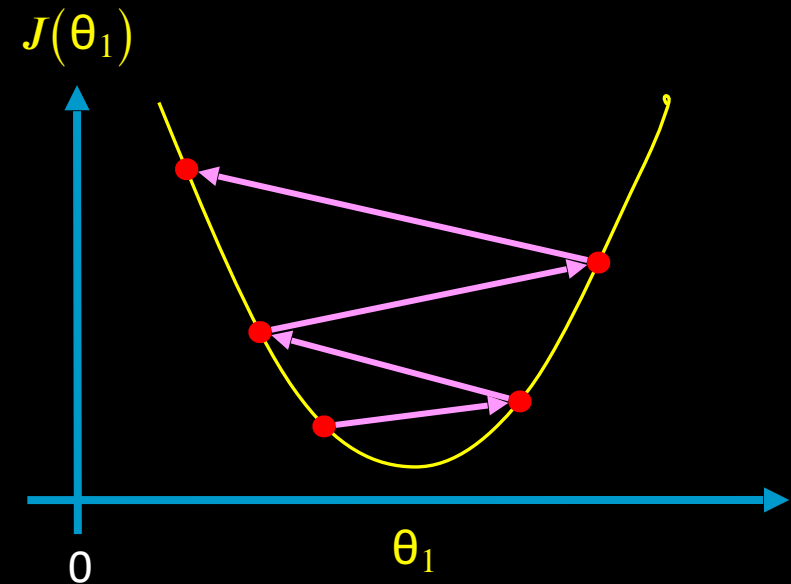
$$\theta_1 = \theta_1 - \alpha \cdot (\text{Negative value})$$

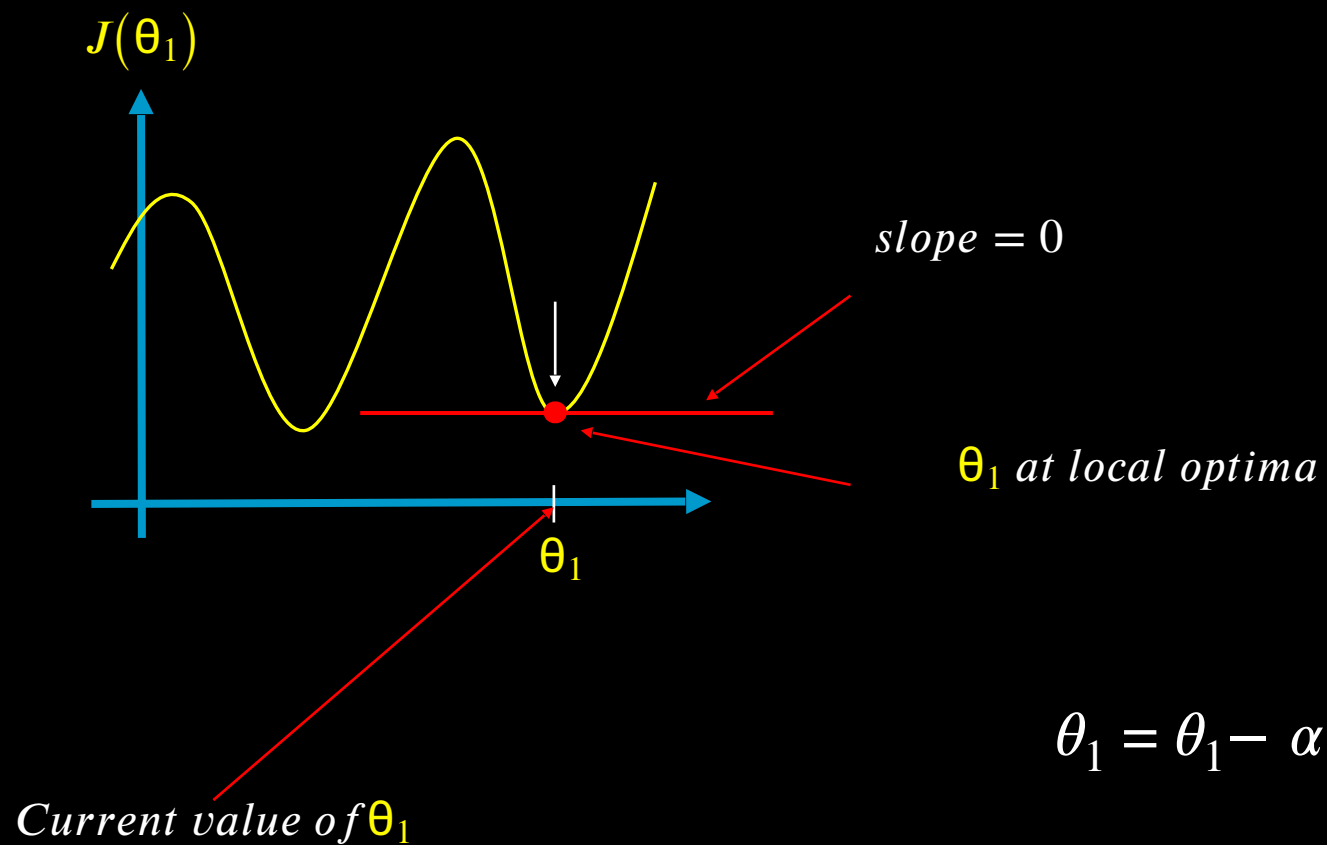
$$\theta_1 = \theta_1 - \alpha \cdot \frac{\partial}{\partial \theta_1} J(\theta_1)$$

If α is too small, gradient descent *can be slow*.



If α is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.





$$\theta_1 = \theta_1 - \alpha \cdot \frac{\partial}{\partial \theta_1} J(\theta_1)$$

$= 0$

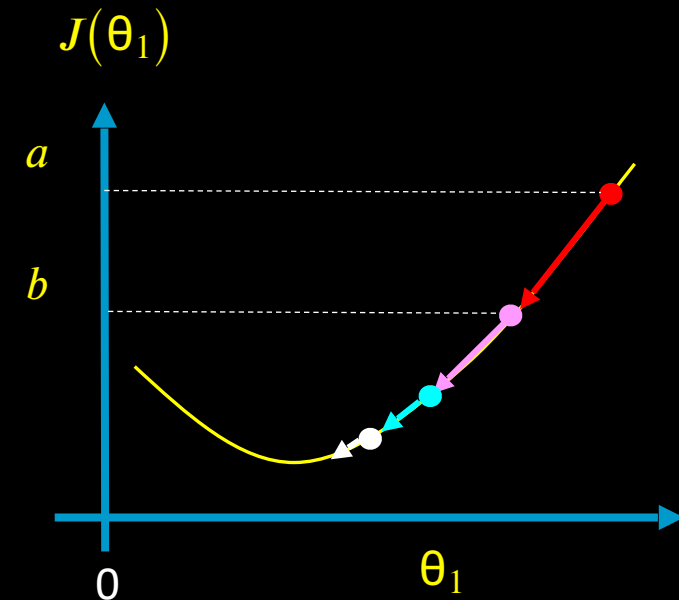
$$\theta_1 = \theta_1 - \alpha \cdot 0$$

$$\theta_1 = \theta_1$$

Gradient descent can converge to a local minimum, even with the learning rate α fixed.

$$\theta_1 = \theta_1 - \alpha \cdot \frac{\partial}{\partial \theta_1} J(\theta_1)$$

As we approach a local minimum, gradient descent will automatically take smaller steps. So, no need to decrease α over time.



Temperature (Dataset)

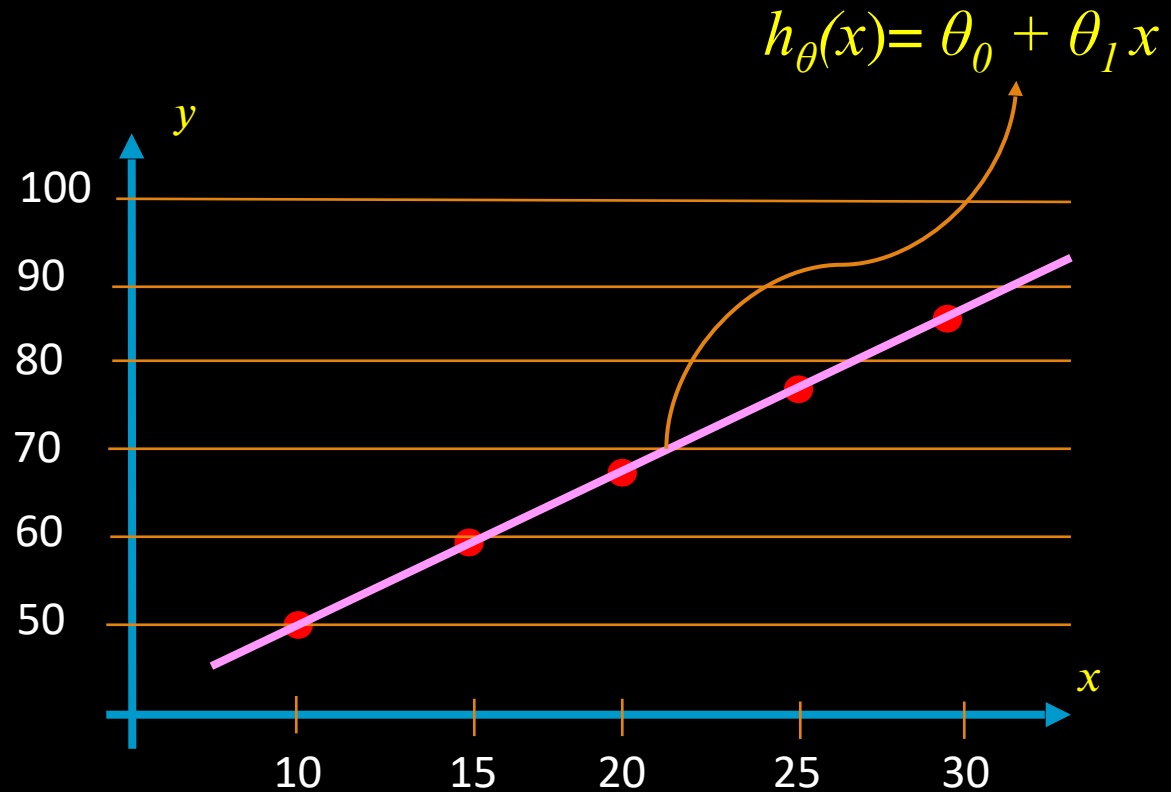
Temp (°C) (x)	Temp (F) (y)
10	50
15	59
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25	77
30	86

$$F = 32 + 1.8 \cdot t$$

$$h_{\theta}(x) = 32 + 1.8 \cdot x$$

$$\theta_0 = 32$$

$$\theta_1 = 1.8$$



Gradient descent algorithm

Repeat until convergence {

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

(for $j = 0$ and $j = 1$)

}

Linear Regression Model

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

$$\underset{\theta_0, \theta_1}{\text{minimize}} \quad J(\theta_0, \theta_1)$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2 = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m \left(\theta_0 + \theta_1 x^{(i)} - y^{(i)} \right)^2$$

$$j = 0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)$$

$$\Rightarrow \theta_0$$

$$j = 1 \Rightarrow \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)}$$

$$\Rightarrow \theta_1$$

Gradient descent algorithm

Repeat until convergence {

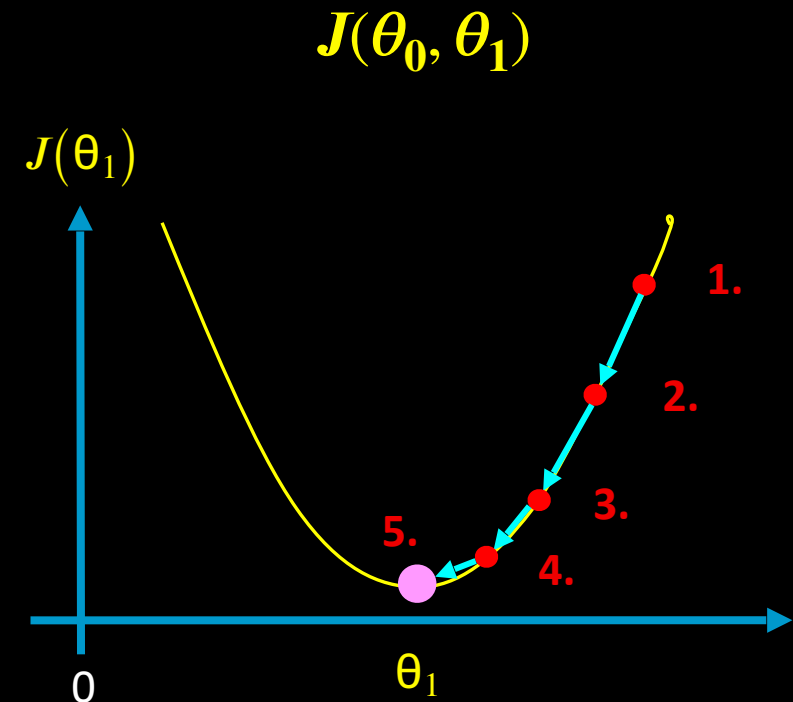
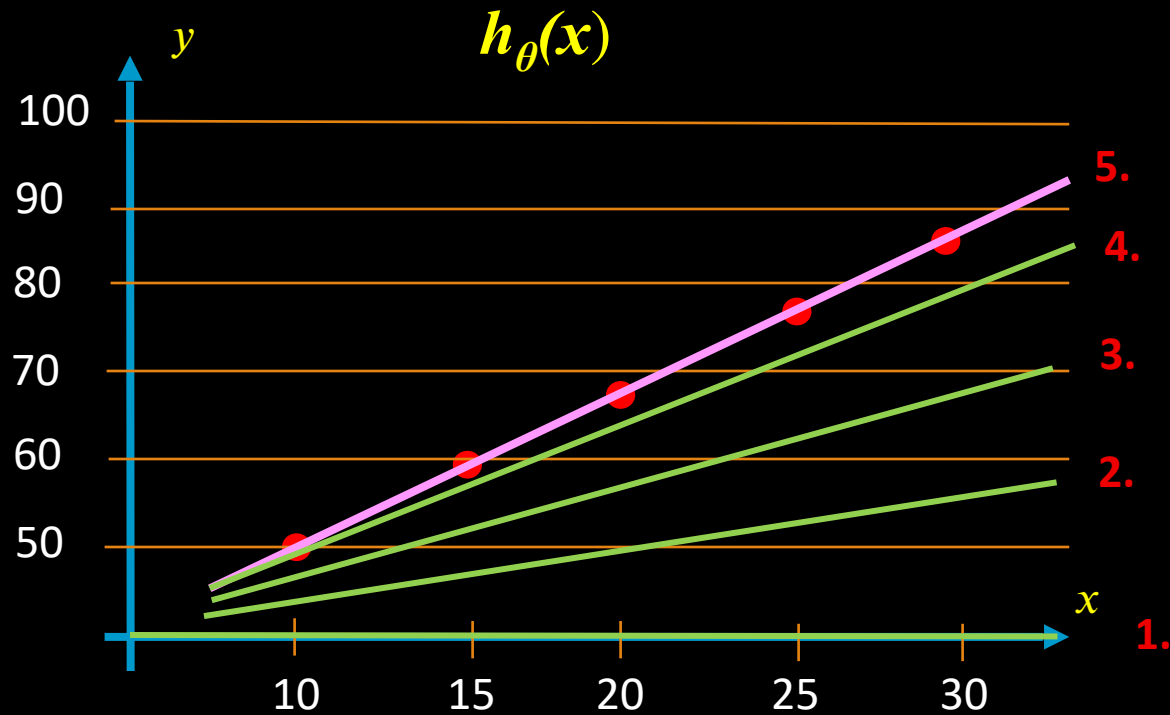
$$\theta_0 = \theta_0 - \alpha \cdot \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$\theta_1 = \theta_1 - \alpha \cdot \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

(for $j = 0$ and $j = 1$)

}

Update
 θ_0 and θ_1



Iterations:

1. $\theta_0 = 0, \theta_1 = 0$
2. $\theta_0 = \dots, \theta_1 = \dots$
3. $\theta_0 = \dots, \theta_1 = \dots$
4. $\theta_0 = \dots, \theta_1 = \dots$
5. $\theta_0 = 32, \theta_1 = 1.8$

Cost function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

Gradient descent:

$$\theta_0 = \theta_0 - \alpha \cdot \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)$$

$$\theta_1 = \theta_1 - \alpha \cdot \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)}$$

Update
 θ_0 and θ_1



